

# Google Places NearbySearch — Root Cause Analysis

**Subject:** ~38,236 `google.places.NearbySearch.Http` requests over 6 days, incl. a ~21,702-request spike on **2026-07-05**.

**Status:** Investigation complete. Root cause proven at the mechanism/per-unit level; exact request-count reconstruction partially proven (see *Remaining Unknowns*). **Constraint honored:** No production code modified. All measurement done offline with a network backstop; **zero live Google requests were issued by this investigation.**

## 1. Executive Summary

The requests were **not** produced by production user traffic, a real MLS import, a queue worker, the scheduler, or browser page views. They were produced by the **automated PHPUnit test suite executing the real Location DNA POI pipeline against the live Google Places endpoint.**

Four independent conditions combined to let test code make live, billable API calls:

1. `QUEUE_CONNECTION=sync` in the test env -> a dispatched `ComputeLocationDna` job runs **inline**, immediately, inside the test.
2. **The `GOOGLE_PLACES_API_KEY` is present in the test process** as a system environment variable (a 39-character value — the standard length of a Google Maps API key; the value itself was not printed). `.env.testing` and `phpunit.xml` do **not** unset it, and `dotenv` cannot override a system env var.
3. **The POI HTTP calls use raw Guzzle (`new Client()`)**, which Laravel's `Http::fake()` **cannot intercept**, and the test base has **no global network guard and no stub binding**.
4. **Almost no feature test isolates the queue:** only **4 of 140** feature-test files call `Bus::fake() / Queue::fake()`. Any of the remaining tests that saves an observer-tracked listing model runs the `observer -> job -> live API chain` synchronously.

Each triggered "generation" makes **exactly 16** `NearbySearch` requests (measured). Because tests use `RefreshDatabase` + an `array` cache, **no caching survives between tests or runs**, so every generation re-fetches all 16 categories from scratch.

### Measured, proven facts (offline, zero live calls):

- Path A (property Location DNA) = **16** `NearbySearch` requests / generation.
- Path B (buyer/tenant lookup) = **7** `NearbySearch` requests / lookup.
- The current suite still *attempts* live `NearbySearch`: a full run made **60** attempts (all intercepted/blocked by the backstop; on 2026-07-05 with the live key these would have reached Google).

**Primary root cause (HIGH confidence):** test-environment isolation failure — live API key + sync queue + un-fakeable raw-Guzzle client + side-effecting observers, with no safeguard at any layer.

**What cannot currently be proven (LOW confidence):** the exact arithmetic reconstruction of 21,702 in a single reproducible run. The current suite yields only ~15 generations/run because reproducing more requires letting blocked external dependencies (Bridge OData, Google Geocoding) succeed — which the backstop deliberately prevents. See §11.

## 2. Timeline

| Date (2026)                               | Event                                                                                                         | Evidence                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 05-28                                     | DNA observers first wired ( <code>PropertyAuctionDnaObserver</code> etc.)                                     | git <code>ca0b7374a</code>                                                     |
| 05-31                                     | Path A caller <code>LocationDnaPoiDistanceService</code> created                                              | git <code>61f63cc74</code>                                                     |
| 06-14                                     | Path B caller <code>GooglePlacesPoiAdapter</code> created                                                     | git <code>f949a7da4</code>                                                     |
| 06-23                                     | Tile cache added (default precision <code>null</code> = <b>disabled</b> )                                     | git <code>bf33bb219</code> ; <code>config/location_dna.php</code>              |
| 06-23...25                                | Bridge lazy-import + "prevent duplicate API calls" work                                                       | git <code>1e09cc255</code> , <code>19c54cf6e</code>                            |
| 06-27 / 06-28                             | Genuine production POI batch (5 <code>seller_agent</code> listings)                                           | <code>property_location_pois</code> created_at: 772 / 193 rows                 |
| <b>07-05 16:51-23:34</b>                  | <b>Intensive Location DNA POI/cache/version dev session</b> (Stage A-E0, provider registry, version stamping) | 15 commits <code>00e606b8b</code> ... <code>12692c6a0</code>                   |
| <b>07-05 (7 bursts, peak 22:51-23:13)</b> | <b>21,702 NearbySearch requests</b>                                                                           | Google Metrics; <code>laravel-2026-07-05.log</code> (323 dispatches, 7 bursts) |
| 07-06                                     | Investigation; API disabled by owner                                                                          | this doc                                                                       |

### 3. Investigation Process

- Enumerated **every** code path that can call `.../place/nearbysearch/json` (exhaustive grep).
- Traced every invoker of the two callers (commands, jobs, observers, controllers, Livewire, bridge services, tests).
- Pulled production DB state ( `heliumdb` ) for persistence, cache, and reusability.
- Parsed the July-5 application log for dispatch source, environment, fixtures, and burst timing.
- Read `.env.testing` / `phpunit.xml` / `tests/TestCase.php` to establish the test-env configuration.
- Measured** per-generation and per-lookup request counts with a counting mock HTTP client (zero network) + transaction rollback.
- Ran the full suite** through a counting HTTPS proxy backstop that intercepts and blocks every outbound request while counting it by host.
- Reconciled measured numbers against the 21,702 / 38,236 metrics.

**Backstop design (safety):** a local CONNECT proxy on `127.0.0.1:8099`; `HTTPS_PROXY` / `HTTP_PROXY` routed all Guzzle traffic to it; it counts each request by host and closes the tunnel so **nothing egresses**. Verified: a direct Guzzle call to the `NearbySearch` URL returned `cURL error 56: Proxy CONNECT aborted` and was counted — proving interception. Plus a fake key, plus the owner having disabled the API = three independent safety layers.

### 4. Hypotheses Tested

| #  | Hypothesis                                                     | Verdict                                                                        | Evidence for                                                                                                                                                    | Evidence against                                                                                                                                                 |
|----|----------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H1 | Production <b>browser</b> traffic (buyer/tenant views, Path B) | <b>DISPROVEN</b>                                                               | Path B exists (7/view) in 4 controllers                                                                                                                         | App not launched; July-5 log has <b>0</b> local-env LocationDna/compose activity (only landlord-draft Livewire mounts); <b>0</b> production persistence on 07-05 |
| H2 | Real <b>MLS import</b> of 667 properties                       | <b>DISPROVEN</b>                                                               | Importer dispatches per record                                                                                                                                  | <b>0</b> ImportBridgeProperties log lines on 07-05; bridge_dna_records = 0 / 667                                                                                 |
| H3 | <b>Queue worker</b> replay                                     | <b>DISPROVEN (impossible)</b>                                                  | —                                                                                                                                                               | QUEUE_CONNECTION=sync ; no queue:work /Horizon/supervisor anywhere                                                                                               |
| H4 | <b>Scheduler / cron</b>                                        | <b>DISPROVEN (impossible)</b>                                                  | —                                                                                                                                                               | Only offers:expire-pending scheduled; makes no Places call                                                                                                       |
| H5 | ldna:refresh-all / rerank-all / benchmark                      | <b>Ruled out as major</b>                                                      | Commands can call Path A                                                                                                                                        | rerank-all is API-free by design; refresh/backfill bounded by 11 DNA records (≤176); benchmark ≤~880/run; no evidence any ran 07-05                              |
| H6 | <b>Automated test suite</b>                                    | <b>SUPPORTED; mechanism &amp; per-unit PROVEN; full count NOT reproducible</b> | 323 testing.INFO dispatches, 7 bursts, synthetic fixtures; measured 60 live attempts even today; key-leak + sync + raw-Guzzle + unfaked observers all confirmed | Current suite yields only 15 generations/run — see §11 for why the exact 21,702 can't be reproduced now                                                          |

By elimination + direct evidence, H6 is the only surviving source. H1-H5 are each affirmatively excluded by architecture, dataset bounds, or log/DB absence.

## 5. Final Root Cause

The test suite ran the **real** LocationDnaPoiDistanceService::calculateForListing() / GooglePlacesPoiAdapter::search() code against the **live** Google Places NearbySearch endpoint, because the test environment had a live API key, a synchronous queue, an un-fakeable HTTP client, and side-effecting observers — with no isolation, no cache, and no quota at any layer.

### Exact execution path (Path A — the dominant one)

```
php artisan test
(APP_ENV=testing, QUEUE_CONNECTION=sync, DB=sqlite :memory:, real GOOGLE_PLACES_API_KEY in env)
|
▼ a feature test saves a PropertyAuction / SellerAgentAuction / LandlordAuction
  (only 4 of 140 feature files fake the queue; the rest run the job synchronously on save)
|
▼ App\Observers\Ndn\PropertyAuctionDnaObserver::saved()      [no dirty-check]
  -> App\Jobs\ComputeLocationDna::dispatch('seller', id)
|
  QUEUE=sync -> job runs INLINE
▼ App\Jobs\ComputeLocationDna::handle()
  -> App\Services\LocationDna\LocationDnaPipelineRunner::run()
    1. geocode 2. POI 3. summary 4. lifestyle
|
▼ LocationDnaPoiDistanceService::calculateForListing()
  RefreshDatabase -> no cached rows; tile cache disabled -> full fetch
  loop 19 categories – 3 grouped = 16 × fetchRawCandidates()
```

```
|      $client = $this->httpClient ?? new Client();    ← raw Guzzle; Http::fake() cannot see it
▼ GET https://maps.googleapis.com/maps/api/place/nearbysearch/json    ← 16 LIVE requests
```

Bridge-fixture path is identical but reaches POI via `LazyBridgeImportService::...` -> `ComputeLocationDna::dispatch('bridge', id)` (323 such dispatches logged on 07-05). Bridge fixtures carry lat/lng, so geocode short-circuits and POI always runs -> 16 each.

Exact files & functions

- `app/Services/LocationDna/LocationDnaPoiDistanceService.php` -> `calculateForListing()` (L474), `fetchRawCandidates()` (L876) — **the NearbySearch call.**
- `app/Services/LocationDna/GooglePlacesPoiAdapter.php` -> `search()` (L57) — Path B NearbySearch call.
- `app/Jobs/ComputeLocationDna.php` -> `handle()` — runs inline under sync queue.
- `app/Observers/Dna/PropertyAuctionDnaObserver.php` / `LandlordAuctionDnaObserver.php` -> `saved()` — no dirty-check.
- `app/Services/Bridge/LazyBridgeImportService.php` (L150) — bridge dispatch (323× on 07-05).
- `tests/TestCase.php` -> `setUpTraits()` — forces sqlite; **no network guard, no stub binding.**
- `.env.testing`, `phpunit.xml` — set sync/array/sqlite; **do not unset the Places key.**

6. Measured Evidence (the numbers)

All measured **offline** (counting mock/proxy, zero live calls):

| Measurement                                                      | Value                               | Method                                                                                                                          |
|------------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| NearbySearch per Location DNA generation (Path A)                | 16                                  | mock client + <code>calculateForListing('seller_agent', 654)</code> in a rolled-back transaction; status <code>completed</code> |
| NearbySearch per buyer/tenant lookup (Path B)                    | 7                                   | mock client + <code>PoiDistanceLookupService::lookup()</code>                                                                   |
| NearbySearch per observer save (that reaches POI)                | 16                                  | = 1 generation; observers dispatch exactly one job, no dirty-check                                                              |
| NearbySearch per bridge import record (reaches POI)              | 16                                  | bridge fixtures carry coords -> geocode short-circuits -> POI runs                                                              |
| <b>Full current suite</b> (7,643 pass / 356 fail-under-backstop) | <b>60 attempts / 15 generations</b> | counting proxy; all blocked                                                                                                     |
| Bridge test dir alone                                            | 14 attempts                         | counting proxy                                                                                                                  |
| Stellar dir alone                                                | 0 attempts                          | counting proxy (tests fail before dispatch under backstop)                                                                      |

**Per-unit is proven and exact.** The 16-per-generation figure independently matches production `location_dna_poi_run_stats` (`categories_fetched_fresh = 16` on every row).

7. Reconciliation to 21,702 / 38,236

- $21,702 \div 16 = \mathbf{1,356 \text{ generations}}$  on 07-05.
- $38,236 \div 16 = \mathbf{2,390 \text{ generations}}$  over 6 days.
- July-5 log directly evidences **323** bridge generations (->  $\geq 5,168$  requests) **plus** unlogged observer generations (observer dispatches log nothing on success).

**Honest gap:** the current full suite reproduces only **15 generations / 60 attempts**, not 1,356. This is **not** contradictory — it is a limitation of the safe backstop:

- The backstop **blocks the external dependencies** the July-5 generating tests relied on to *proceed* to the POI step — Google Geocoding (for seller/landlord observer saves) and the Bridge OData API (`api.bridgedataoutput.com`, 15 CONNECTs observed). When those calls fail at the proxy, the pipeline stops **before** `NearbySearch`, so the attempts never happen under the harness.
- On 07-05, with the live key and live/faked externals succeeding, those same generations completed and reached the 16-call POI step.

So the mechanism and per-unit cost are proven; the **exact** July-5 per-run count is **not reproducible offline** without letting the external calls succeed (which would require modifying test/production code and risking live calls). See §14.

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## 8. Why the requests repeated

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- **RefreshDatabase resets the DB before every test** -> the per-listing coordinate cache in `property_location_pois` is empty every time -> each generation re-fetches all 16 categories.
- **CACHE\_DRIVER=array** in tests -> the tile cache and Path-B `poi_lookup` cache are in-memory and **discarded per test/run** -> no reuse.
- The suite (and individual Location-DNA tests under active development) was **run many times** during the 07-05 session (7 bursts).

## 9. Why caching failed

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- **Tile cache disabled in production:** `LOCATION_DNA_POI_TILE_PRECISION` unset -> `precision_used = null`, `categories_from_tile_cache = 0` on every run.
- **Tests use array cache** -> nothing persists across tests or runs.
- **RefreshDatabase** wipes the per-listing DB coordinate cache each test.
- Even in production, **1,089 / 1,090 stored POI rows have** `pois_fetch_version = NULL`, which the reader treats as **stale** vs the current `06275e91...` -> they would be re-fetched anyway.

## 10. Why testing hit the live API

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1. **QUEUE\_CONNECTION=sync** ( `.env.testing` ) -> dispatch executes the job inline.
2. **Live key leaks in:** `.env.testing` does not set `GOOGLE_PLACES_API_KEY`; it exists as a **39-char system env var** that `dotenv` cannot override.
3. **Raw Guzzle** ( `new Client()` ) in both POI callers -> `Http::fake()` is powerless.
4. **No global test guard:** `tests/TestCase.php` has no `Http` fake and no stub binding of `PoiLookupAdapterInterface` / `LocationDnaPoiDistanceService`.
5. **Only 4 of 140** feature files fake the queue/bus; the rest execute dispatched jobs synchronously when they save an observer-tracked model.

## 11. Why the API key wasn't blocked in testing

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`.env.testing` sets `sqlite/array/sync` but **omits** `GOOGLE_PLACES_API_KEY`; `phpunit.xml` also omits it. The key is injected as a **system-level secret** (same mechanism `tests/TestCase.php` documents for `DB_CONNECTION`), which `dotenv`'s immutable

repository will not override. **Proof:** `.env.testing` contains no `GOOGLE_PLACES_API_KEY` line; the test process reports the variable `SET` at length 39 (the standard length of a Google Maps API key; the value was not inspected).

## 12. Why monitoring didn't catch it

- The POI callers **log nothing** about outbound requests — no URL, no count, no status. Grep of the 07-05 log for `nearbysearch|maps.googleapis|REQUEST_DENIED|guzzle = 0`.
- There is **no metric/counter/alert** on Places usage in code.
- It was discovered only via the **Google Cloud bill**, days later.

## 13. Why budget controls didn't stop it

- **No code-level quota / budget / circuit-breaker / rate-limit** exists on the Places path (grep confirms none).
- **No Google Cloud budget cap, quota limit, or API-key restriction** was configured (external to this codebase; the owner reports having disabled the Places API after the spike was discovered).

## 14. Remediation Checklist (status)

**Legend:** [DONE] Completed · [PLANNED] Planned / in progress · [TODO] Not yet implemented

As of 2026-07-06, the only completed items are the containment action taken by the owner (disabling the API) and this investigation. **No source-code fixes have been implemented** — code changes were paused pending review. The statuses below are honest and auditable; move items to [PLANNED]/[DONE] as they are scheduled and shipped.

### Containment (already done)

| Status | Action                                                                 | Evidence                    |
|--------|------------------------------------------------------------------------|-----------------------------|
| [DONE] | <b>Google Places API disabled</b> in the affected Google Cloud project | Owner-reported (2026-07-06) |
| [DONE] | <b>Root-cause investigation</b> completed with offline measurement     | This document               |

### Prevent live calls from tests (highest priority)

| Status | Action                                                                                                                                                                                                                    | Owner | Notes                                            |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------|
| [TODO] | Bind <code>PoiLookupAdapterInterface</code> -> <code>StubPoiLookupAdapter</code> <b>and</b> bind <code>LocationDnaPoiDistanceService</code> to a no-network instance in the test base ( <code>tests/TestCase.php</code> ) | —     | Stops the observer->job->API chain in every test |
| [TODO] | Set <code>GOOGLE_PLACES_API_KEY=</code> (empty) in <code>.env.testing</code> and <code>phpunit.xml</code> ; add a guard test asserting the key is blank under <code>APP_ENV=testing</code>                                | —     | Code already no-ops on a blank key               |
| [TODO] | Resolve the HTTP client from the container in both POI callers (remove bare <code>new Client()</code> ) so faking works by default                                                                                        | —     | Makes <code>Http</code> /mock faking effective   |
| [TODO] | Add a global stray-request guard so any un-mocked outbound call in tests fails loudly                                                                                                                                     | —     | Fail-closed instead of fail-to-network           |

### Reduce production blast radius

| Status | Action                                                                                                                         | Owner | Notes                                          |
|--------|--------------------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------|
| [TODO] | Add a <b>dirty-check</b> to the DNA observers (dispatch only when address/coords change)                                       | —     | Removes redundant regenerations                |
| [TODO] | Add a <b>code-level circuit breaker</b> / <b>daily request cap</b> around the NearbySearch callers                             | —     | Hard ceiling regardless of caller              |
| [TODO] | <b>Enable the tile cache</b> ( <code>LOCATION_DNA_POI_TILE_PRECISION</code> ) and fix <code>pois_fetch_version</code> stamping | —     | Currently disabled; 1,089/1,090 rows are stale |
| [TODO] | <b>Log/emit a metric</b> for every Places request (endpoint, listing, count)                                                   | —     | No outbound-call observability today           |

External (Google Cloud console)

| Status | Action                                                | Owner | Notes              |
|--------|-------------------------------------------------------|-------|--------------------|
| [DONE] | Disable the API (containment)                         | —     | Done; see above    |
| [TODO] | Restrict the API key (API scope + HTTP referrer / IP) | —     | Before re-enabling |
| [TODO] | Set a <b>daily quota cap</b> on NearbySearch          | —     | Before re-enabling |
| [TODO] | Set a <b>budget alert / threshold</b>                 | —     | Before re-enabling |

15. Remaining Unknowns (cannot currently be proven)

- **The exact tests and run-count that produced the 1,356 generations on 07-05.** The POI code logs no HTTP, and the 273 `md5`-style bulk fixture keys from the 22:51–23:13 burst could not be pinned to a specific current test. *Cannot be proven from available evidence.*
- **The precise per-run July-5 request count** — not reproducible offline because the safe backstop blocks the external dependencies (Bridge OData, Geocoding) the generating tests need to proceed. *Provable only by (a) re-running the exact 07-05 tests with faked externals + a counting client — a code change — or (b) Google Cloud per-minute metrics overlaid on the 7 burst windows.*
- **Billable vs REQUEST\_DENIED split.** Tests that set a fake key would return HTTP 200 `REQUEST_DENIED` (counts as a 2xx "request" in Metrics but is typically unbilled); tests using the leaked real key return billable `OK / ZERO_RESULTS` . Only Google's per-response metrics can split the 21,702 into billed vs unbilled.

16. Confidence Levels

| Conclusion                                                                     | Confidence | Basis                                                              |
|--------------------------------------------------------------------------------|------------|--------------------------------------------------------------------|
| Per generation = 16 NearbySearch; per lookup = 7                               | PROVEN     | Direct offline measurement + matches <code>run_stats</code>        |
| Tests attempt live NearbySearch calls                                          | PROVEN     | 60 intercepted attempts in a current full run                      |
| Live key leaks into the test env                                               | PROVEN     | <code>.env.testing</code> omits it; 39-char system env var present |
| Root cause = test suite via sync + leaked key + raw Guzzle + unfaked observers | HIGH       | All four conditions confirmed; only surviving hypothesis           |
| Not production / import / cron / queue / browser                               | HIGH       | Each affirmatively disproven (log/DB/architecture)                 |
| Caching absent/ineffective; no safeguards; no monitoring                       | PROVEN     | Config + grep + DB state                                           |
| Exact reconstruction of 21,702 in a single run                                 | LOW        | Not reproducible offline; documented why (§7, §15)                 |

Prepared 2026-07-06. No production code was modified. Measurement artifacts (counting proxy, mock-client harness, DB probes) are in the session scratchpad and can be re-run on request.